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PEDIATRICS

New instrument for quantitative measurements of passive duction forces and its clinical implications

Hyunkyoo Kang[1] · Shin-Hyo Lee[2] · Hyun Jin Shin[3,4] · Andrew G. Lee[4,5,6,7,8,9,10,11]

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Abstract

Purpose Evaluating the passive duction force of the extraocular muscles is important for the diagnosis of and surgical planning for strabismus. This is especially relevant in patients with an observable limitation of duction movement. The purpose of this study was to validate passive duction forces in healthy subjects using a novel instrument.

Methods An instrument for making continuous quantitative measurements of passive duction forces was designed. Tension was measured as the eyeball was rotated horizontally or vertically from the resting position under general anesthesia 10 mm (50°) away from the direction of force to be tested (opposite side).

Results Seventy eyes of 35 subjects were enrolled in this study (age range of 4–80 years and mean age of 36.3 years). The passive duction force was measured at 49.0 ± 15.3 g (mean ± standard deviation) for medial rotation, 44.8 ± 13.2 g for lateral rotation, 50.5 ± 14.8 g for superior rotation, and 53.5 ± 13.8 g for inferior rotation. The passive duction forces were similar for all gaze positions, but it was larger for inferior rotation than for lateral rotation ($P = 0.009$). The passive duction force was significantly larger for vertical rotation (51.9 ± 14.4 g) than for horizontal rotation (46.9 ± 14.4 g) ($P = 0.006$). The passive duction force did not differ significantly with sex ($P = 0.355$), side ($P = 0.087$), or age ($P = 0.872$).

Conclusions These measurements of passive duction forces in a healthy population provide valuable information for diagnosing specific strabismic problems and could be useful for increasing the precision of strabismus surgery.

Keywords Extraocular muscle · Forced duction · Passive duction force · Strabismus · Tension

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✉ Hyun Jin Shin
shineye@kuh.ac.kr

1 Department of Mechatronics and Electronic Engineering, Konkuk University Glocal Campus, Chungcheongbuk-do, Republic of Korea

2 Department of Anatomy, Yonsei University College of Medicine, Seoul, Republic of Korea

3 Department of Ophthalmology, Research Institute of Medical Science, Konkuk University Medical Center, Konkuk University School of Medicine, 120 Neungdong-ro, Gwangjin-gu, Seoul 05030, Republic of Korea

4 Department of Ophthalmology, Blanton Eye Institute, Houston Methodist Hospital, Houston, TX, USA

5 Department of Ophthalmology, Neurology, and Neurosurgery, Weill Cornell Medicine, New York, NY, USA

6 Department of Ophthalmology, University of Texas Medical Branch, Galveston, TX, USA

7 Department of Ophthalmology, UT MD Anderson Cancer Center, Houston, TX, USA

8 Department of Ophthalmology, Texas A&M College of Medicine, College Station, TX, USA

9 Department of Ophthalmology, University of Iowa Hospitals and Clinics, Iowa City, IA, USA

10 Department of Ophthalmology, Baylor College of Medicine and the Center for Space Medicine, Houston, TX, USA

11 Department of Ophthalmology, University of Buffalo, Buffalo, NY, USA

Key Messages

- The forced duction test (FDT) is a simple and easy method for clinically evaluating the mechanical properties of the extraocular muscles (EOMs). However, the results from the FDT are highly dependent on the experience and skill of clinicians, and also this test cannot be used to detect small pathological changes.

- A method for quantitatively analyzing the passive tensions in EOMs is required to allow more accurate and objective assessments. Thus, the authors have designed a simple and compact device for quantitatively and continuously measuring the passive duction force in a EOMs.

- The passive duction force measured in the four fields was 49.4±14.6 g (0.84±0.27 g/deg), and was similar for all gaze positions. The values reported here could provide the baseline for passive forced duction testing of healthy eyes and aid clinicians in making differential diagnoses of restrictive or paralytic strabismus and result in more accurate quantitative planning of strabismus surgery.

Introduction

The management of strabismus requires knowledge of the mechanical and innervational properties of the extraocular muscles (EOMs), as well as of the mechanical and viscous properties of the orbital passive, restraining tissue [1, 2]. This knowledge is especially important in the diagnosis and surgical treatment of patients with an observable limitation of ductional movement. In this setting, the ophthalmologist must determine whether a rotational limitation is due to muscle paresis, mechanical restriction, or both conditions.

The usefulness of passive forced duction testing has been emphasized in the literature for the past decades [3–5]. This clinical test is usually performed based on the subjective and qualitative assessment of resistance perceived by the hand of the ophthalmic surgeon while rotating the eyeball with forceps applied to the limbus. The passive forced duction test however even in expert hands is subjective, and this technique alone usually cannot detect small pathological changes. In addition, there is significant intra-rater and inter-rater reliability for the test [6, 7]. The qualitative nature of the testing can lead to inaccuracy and variability in assessment of the passive tension and might adversely affect the strabismus surgical management [4].

Although several currently available quantitative instruments and recording systems exist for extraocular tension measurements, most are confined to the research laboratory setting and are too expensive, complicated, or complex for use in the clinic or operating room [8–10]. We have therefore developed a novel device for performing quantitative measurements and have simplified the technique so that it can be easily applied in clinical practice. The aim of this study was to present quantitative baseline data of the passive forces acting on a healthy eye by compression weighing of force sensors. These data may aid clinicians in making differential diagnoses, resulting in more accurate planning for strabismus management.

Materials and methods

This prospective study was conducted at the Department of Ophthalmology, Konkuk University Medical Center, Seoul, Korea, between May 2019 and February 2020. The study protocol complied with the Declaration of Helsinki and was approved by the Institutional Review Board (IRB) of Konkuk University Medical Center (registration number: KUH1100071). Informed consent was obtained from all of the included participants.

All participants were within 2Δ of being orthophoric without limitation of eye movement on duction or version testing. Exclusion criteria were the presence of other ocular diseases (e.g., prior history of strabismus, nystagmus, ptosis, orbital diseases, blepharospasm, or glaucoma), thyroid disorder, muscular or neurological diseases (e.g., Parkinson or myasthenia gravis), history of ocular trauma, previous ocular or periocular surgery, history of receiving medications known to affect muscle tension within 3 months (e.g., muscle relaxant or botulinum toxin), or eyes with spherical equivalents of > 3.00 or < − 3.00 diopter.

Quantitative device for passive forced duction measurements

We have developed a novel compact device for quantitative passive forced duction testing (Fig. 1) based on a miniaturized compressive-type load cell (EX-601D-10, Caskorea, Gyeonggi-do, Korea), a biaxial tilt sensor (SA2, DAS, Gyeonggi-do, Korea), and Castroviejo locking forceps (Storz E-1798-S, Karl Storz, Tuttlingen, Germany). Data collection and storage were performed using a data acquisition board (USB-6001, National Instruments, Austin, TX, USA), amplifier (AL-50, Caskorea, Gyeonggi-do, Korea), and laptop. The amplified analog passive duction force signal as well as the two tilt angles were converted and stored in the laptop in digital form using the commercially available LabVIEW software (National Instruments). The passive duction force signal and the tilt angles were displayed on an LCD monitor in both the moving and perpendicular directions. This setup enables the surgeon to maintain an appropriate posture by carefully moving the measuring unit.

Passive duction force measurement procedure

All participants were confirmed as being at American Society of Anesthesiologists (ASA) physical status classification I and were scheduled to receive elective oculoplastic surgery (epiblepharon correction or dacryocystorhinostomy) under general anesthesia. A standardized anesthetic regimen was applied to all patients by a single anesthesiologist to produce a consistent depth of anesthesia. All patients arrived at the operation room without anticholinergic premedication. Anesthesia was induced by administering 5 mg/kg thiopental sodium and 0.6 mg/kg rocuronium (non-depolarizing muscle relaxant). After performing tracheal intubation, anesthesia was maintained by administering sevoflurane with 40% oxygen in air.

The eyeball was moved by applying external forces, and so the tensions were generated in a passive state. Four sets of measurements were made on each eye under general anesthesia, as described previously [1, 11]. Castroviejo locking forceps were attached to the eyeball just posterior to the limbus perpendicularly at the 3-, 6-, 9-, and 12-o'clock positions (Fig. 2A). Medial, lateral, superior, and inferior rotations (away from the direction of force to be tested) were performed, and a counterforce was then generated above the load cell. To compensate for any changes in the eye position before the measurement of tension, if the eyes were not in their primary position, passive duction forces were measured after they were placed in the primary position. Also, calibration was done for the sensing unit and measurement device was zeroed every time just after attaching the Castroviejo locking forceps to the

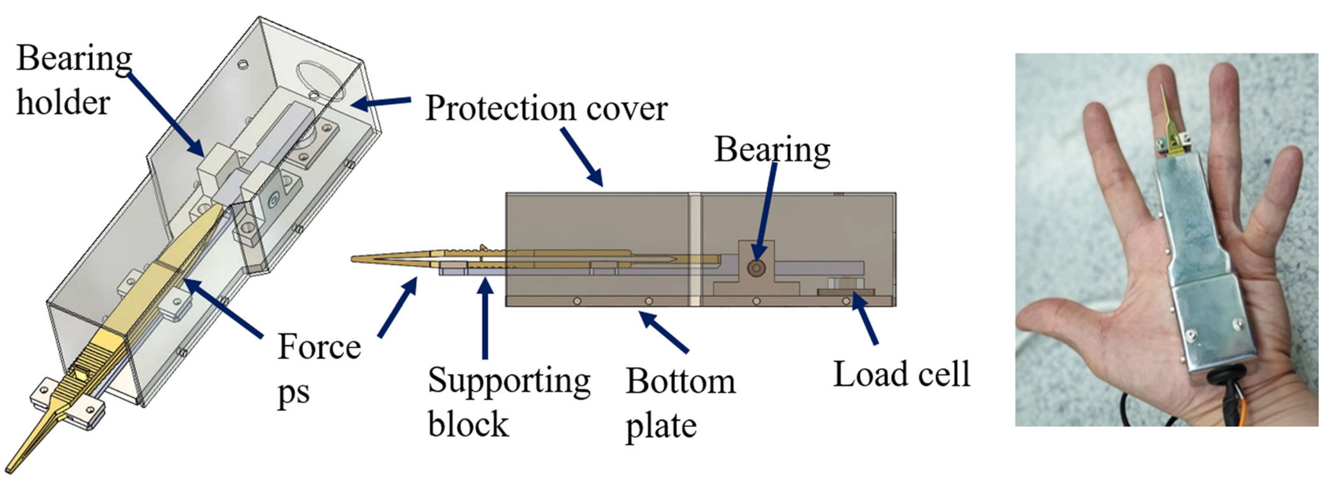


Fig. 1 A novel instrument for quantitative measurements of passive duction forces based on compression weighing of force sensors (load cell)

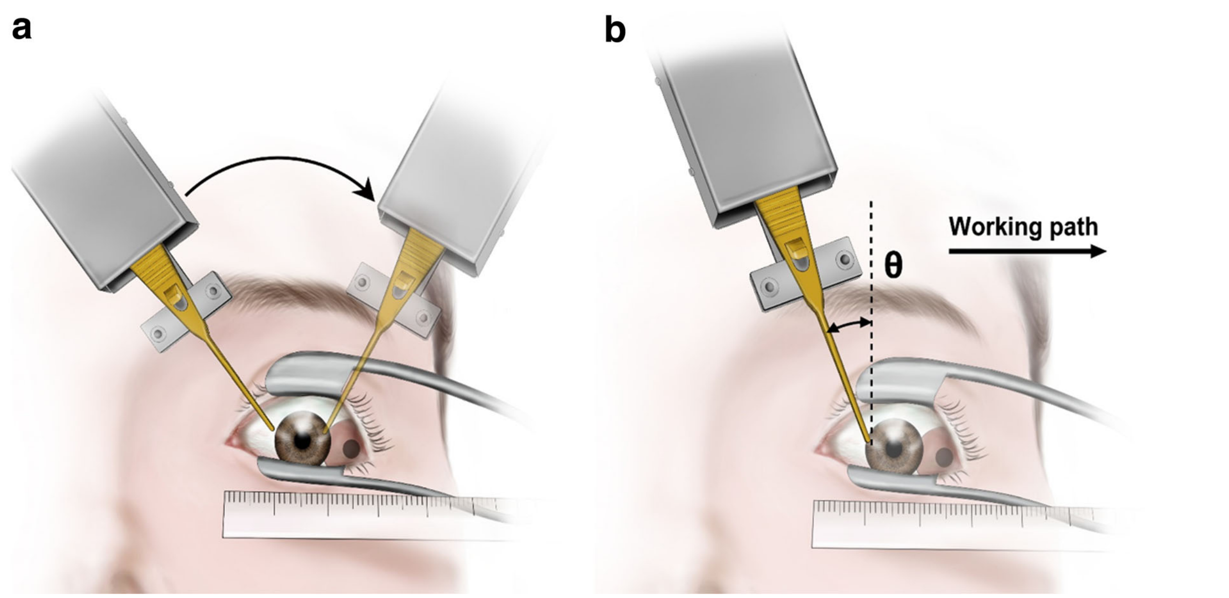


Fig. 2 Schematic illustrations of measuring the passive duction forces. **A** Locking forceps were attached to the limbus and rotated away from the direction of force to be tested. **B** The titling angle (θ) was measured simultaneously, with the calculated passive duction force adjusted according to the device tilt angle

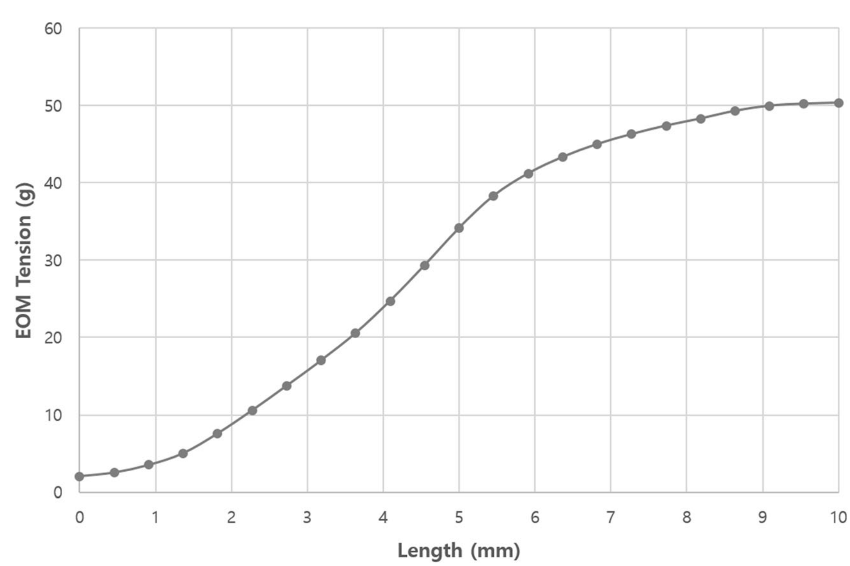
globe and before rotating the forceps. The measurements were recorded after inducing 10 mm (50°) of rotation from the passive position of the eyeball into each of the four fields. Distances were measured by visually aligning the limbus with a ruler.

The employed load cells measure the perpendicular force. The measured tilt angle (θ) was used to compensate for the angle at which the passive duction force was being applied to the load cell (Fig. 2B). The correlation between the tilting angle and applied force was formulated as $T = T_p / \cos(\theta)$, where T is the applied passive duction force, T_p is the force measured in the perpendicular direction, and θ is the tilt angle of the sensing unit. In addition, based on the assumptions that (i) the eyeball is a perfect sphere, (ii) the rotating center of the eyeball is fixed, and (iii) the mean ocular diameter is 25 mm, we calculated the angle of eyeball rotation from the eyeball moving distance as $\theta_r = 2\sin^{-1}(d/D)$, where θ_r is the angle of eyeball rotation, d is the eyeball moving distance, and D is the diameter of the eyeball [12, 13].

Statistical analyses

All calculations and statistical analyses were performed using SPSS for Windows (version 18.0, SPSS, Chicago, IL, USA). The Shapiro–Wilk test was used to determine whether the data followed a parametric (Gaussian) or nonparametric (non-Gaussian) distribution. ANOVA with post-hoc analysis was used for comparisons among the four fields. Sex differences

Fig. 3 Representative recording of the length–tension relationship for medial rotation of the right eye, demonstrating an S-shaped curve. The patterns were similar in the other directions



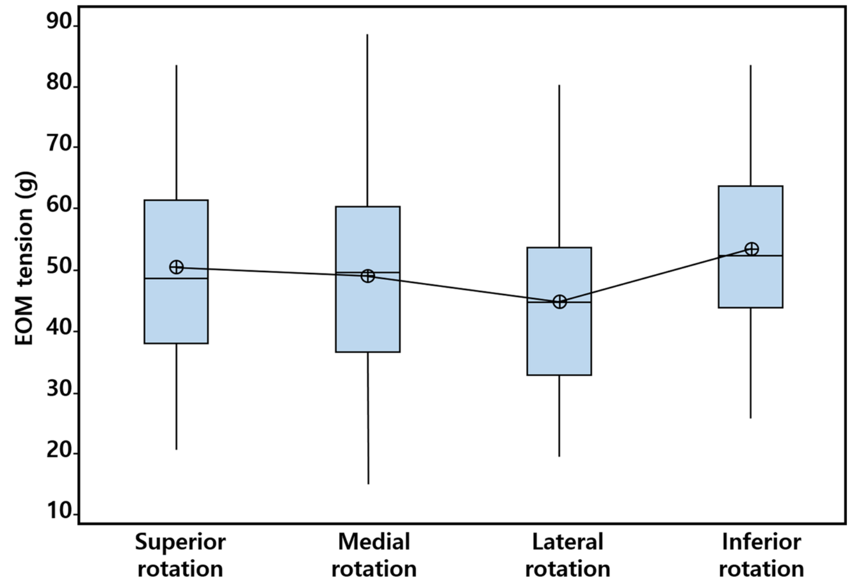


Fig. 4 Box plots of the passive duction forces in the four fields. Each box plot shows the median, first and third quartiles, and range

were analyzed using the Mann–Whitney test, while the side differences (laterality) were analyzed using the paired-samples *t* test. The linearity of relationships between the study parameters was evaluated using Pearson's correlation test. The criterion for statistical significance was set at $P < 0.05$.

Results

This study included 35 healthy subjects with a mean age of 36.3 years (age range of 4–80 years; 9 males, 26 females). The spherical-equivalent refractive error of the enrolled subjects was − 0.09 ± 0.80 diopter (mean ± standard deviation) in the right eye and –0.05 ± 0.84 diopter in the left eye. Both eyes were measured in 29 subjects (58 eyes), and only one eye was measured in 6 of the 35 healthy subjects. A total of 64 eyes were included in the study. The passive duction forces required to move the eye 10 mm (50°) medially, laterally, superiorly, and inferiorly from the primary position were measured. All of the data conformed to a Gaussian distribution in the Shapiro–Wilk test. The passive duction force in the four fields was 49.4 ± 14.6 g, corresponding to 0.85 ± 0.27 g/deg. The continuous measurements performed during the passive forced duction testing produced an S-shaped curve (Fig. 3).

Figure 4 shows the passive duction force in each field, which was 49.0 ± 15.3 g (0.93 ± 0.34 g/deg) for medial rotation, 44.8 ± 13.2 g (0.85 ± 0.27 g/deg) for lateral rotation, 50.5 ± 14.8 g (0.78 ± 0.23 g/deg) for superior rotation, and 53.5 ± 13.8 g (0.82±0.21 g/deg) for inferior rotation. The passive duction forces were similar for all gaze positions in the present

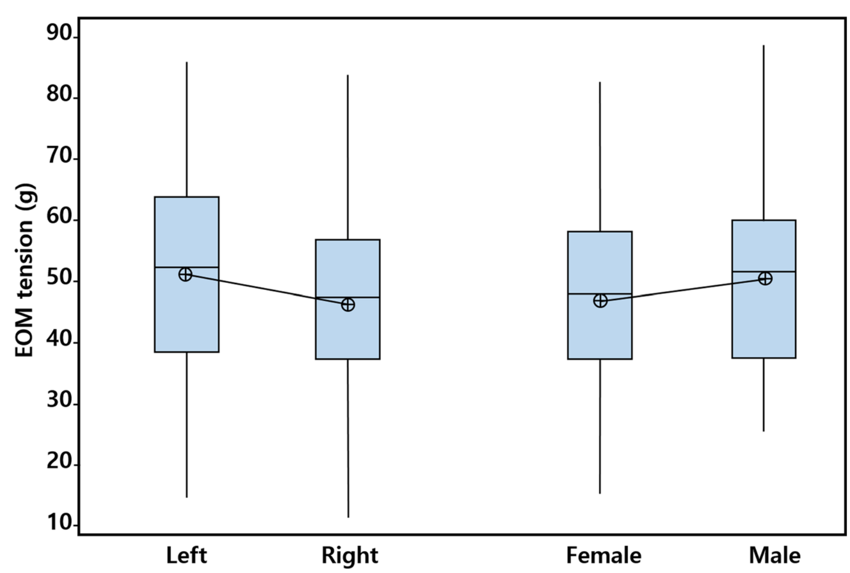


Fig. 5 Box plots of the passive duction forces in the four fields for left and right eyes (right) and females and males (left)

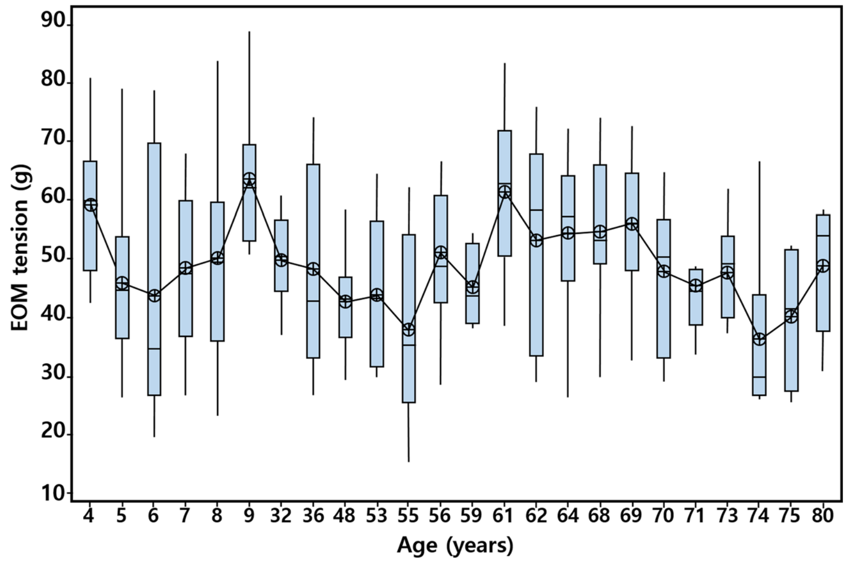


Fig. 6 Box plots of the passive duction forces in the four fields according to age

healthy population, but it was larger for inferior rotation than for lateral rotation ($P = 0.009$). The passive duction force was significantly larger for vertical (inferior and superior) rotation (51.9 ± 14.4 g) than for horizontal (medial and lateral) rotation (46.9 ± 14.4 g) ($P = 0.006$).

In the subgroup analysis, the passive duction force in the four fields was 51.3 ± 15.2 g in males and 48.8 ± 14.4 g in females, and 47.9 ± 13.9 g on the right side and 51.1 ± 15.1 g on the left side. There was no sex difference ($P = 0.355$) or side difference ($P = 0.087$) in the same individual (Fig. 5). Figure 6 demonstrates how the passive duction force varied with age. The regression model showed that there was a tendency for the passive duction force to decrease in those older than 70 years, but there was no significant correlation between age and passive duction force in the four fields (Pearson correlation coefficient = − 0.01, $P = 0.872$).

Discussion

Passive forced duction testing has proven important for evaluating strabismus patients and planning strabismus surgery [3–5]. This testing procedure involves the surgeon rotating the eyeball of the patient with forceps to manually assess the tightness but is subjective and qualitative. Although experienced strabismologists can usually interpret the results of such qualitative forced duction measurements, many general ophthalmologists and less experienced strabismus surgeons may require better instruments to supply quantitative information in order to ensure that reliable results are obtained [6, 7]. Several previous investigators have developed methods to quantify the passive duction force. Stephens and Reinecke [14] attached to a force displacement transducer to a perilimbal suction cup that was placed on the eyeball. Schillinger [15] attached a spring tension gauge to scleral sutures to compare the ease of abduction and adduction when a constant force was applied. Rosenbaum and Mye [11] used a spring dynamometer that was attached to limbal conjunctiva via self-retaining Pierce forceps. However, all of the instruments developed previously for this purpose have been difficult, invasive, and time-consuming to use, and none of these methods are available for application in clinical practice.

The present study was motivated by a desire to enable both beginner and expert strabismologists to measure the passive duction force easily and accurately. We have developed a novel device for quantitative measurements of passive duction forces, with its design simplicity and compact size facilitating its clinical application. The measurement process includes the use of locking forceps that can be attached to the limbus of the conjunctiva. This method is noninvasive and similar to the method of conventional forced duction testing with forceps, making it easy to perform and so lowering the barrier to application by clinicians.

The passive duction force measured in the four fields was 49.4 ± 14.6 g (0.84 ± 0.27 g/deg) and was similar for all gaze positions. The values reported here could provide the baseline for passive forced duction testing of healthy eyes and aid clinicians in making differential diagnoses of restrictive or paralytic strabismus and result in more accurate quantitative planning of strabismus surgery. For example, the maximum passive duction force may be useful when diagnosing restrictive strabismus. Since we found that the passive duction force of the normal EOMs was 49.4 ± 14.6 g, a maximum tension force of more than 78.6 g (i.e., the mean value plus two standard deviations, corresponding to the upper limit of the 97% confidence interval) could lead to suspicion of muscle restriction. It was suggested previously that the force measured under local anesthesia should remain below 100 g in healthy individuals [14, 16, 17].

Stephens and Reinecke [14] reported that the passive duction forces for medial and lateral rotations measured under local anesthesia were 67 and 65 g, respectively, which are larger than our values of 49.0 and 44.8 g. In the present preliminary study we applied general anesthesia when analyzing the passive duction force in a controlled situation. Most patients experience some level of anxiety or discomfort under local anesthesia [18], and afferent emotional or painful stimuli might also be related to muscle tension [19]. However, these subjective factors are difficult to measure and differ between individuals and could be confounding factors. Thus, normative data on passive duction forces obtained under local anesthesia might be less reliable than those measured when the patient is fully anesthetized. The passive duction force could be overestimated under local anesthesia, since the patient anxiety, discomfort, and pain could increase the tension in EOMs.

One strength of our device is the use of a tilting sensor, which has not been used in previous studies. The passive tension forces of the EOMs were measured using horizontal or vertical rotation, while the device was attached to the limbus of the conjunctiva. Although a clinician is able to grasp the limbus perpendicularly, the angle between the device and ocular surface could change during the measuring movement. A tilting sensor for measuring the angle is attached to compensate for the change in force according to the tilt in our device, which improves the accuracy of the tension measurements. We also used locking forceps to rule out any effect of the gripping force.

However, the passive duction force varied between the individual healthy subjects in this study, which suggests that care should be taken when interpreting the results. In this study the passive duction force was measured indirectly by grasping the conjunctiva, rather than the EOM directly. This evaluation method has the advantage of measuring the passive duction force easily, noninvasively, and rapidly. However, in some instances these indirect measures could be affected by the status of supporting tissue surrounding the EOM, such as the conjunctiva and Tenon's capsule. The tension in EOMs measured using an indirect method could be increased if there are any mechanical restrictions in the connective tissue, such as due to conjunctival scarring. Also, this indirect method of rotating the eyeball by grasping the conjunctiva might result in inaccurate measurements in aged subjects who have a stretched conjunctiva and loose attachment from the underlying sclera.

The continuous recording of passive duction forces produced an S-shaped tension–length curve. The shape of the eye position and passive duction force curve leads us to an understanding of otherwise hidden mechanisms of strabismus [20]. For example, in the absolute restriction type (e.g., Graves' ophthalmopathy), the passive duction force could increase immediately when forceps are applied with a small passive forced duction movement. In a uniform type of restriction, the passive duction force tends to increase linearly due to the progressive increase in resistance during the passive traction across the arc of rotation; examples of this type include restrictions secondary to scar tissue and muscle contracture.

The effect of age on the passive duction force is depicted in Fig. 6. This force appeared to be lower in participants who were older than 70 years, but there were no significant age-related differences. It could be considered that the passive duction forces of EOMs are relatively well maintained regardless of aging and that these muscles can work properly even in the elderly without decreases in tension. These observations are consistent with the findings of the histochemical study of Kallestad et al. [21], which suggested that EOMs continuously remodel throughout the lifetime to maintain a population of activated satellite cells even during aging, and thus being spared from the usual pathology associated with aging and many forms of muscular dystrophy.

We recognize that this study has some limitations. First, the sample was relatively small (n = 64 eyes). Second, the measurements of passive duction force were performed in only in four directions, while it would be important to additionally measure the vertical forced duction in abduction and in adduction. Third, we did not measure the active duction forces of the EOMs or the rotatory passive duction forces for oblique muscles, which are generally known to be smaller than the vertical and horizontal forces. The active duction force could be measured in the same manner under local anesthesia by asking the participant to maintain a certain gaze position. Fourth, we calculated the value based on the assumption of an ocular diameter of 25 mm, since we did not measure the actual diameter of the eyeball in all subjects. However, to exclude patients with very long or very short eyeballs, we excluded eyes with spherical equivalents of > 3 and < − 3 diopter. Lastly, the study was confined to healthy subjects without strabismus. Despite these limitations, we believe that our instrument may provide easy to use quantitative measures of forced duction. Future studies will be needed to measure the passive duction forces in patients with various types of strabismus and compare the measurements with the present normative data.

In conclusion, human passive duction forces have been measured using noninvasive techniques by means of load transducers connected via forceps to the limbus of the conjunctiva. These measurements of passive duction forces have provided useful novel information about the physiology and pathology of EOMs. We believe that our instrument for making continuous quantitative measurements will assist in increasing the accuracy of strabismus diagnoses and be helpful in therapeutic decision-making related to therapy timing and type.

Authors' contributions HK Kang developed instrument and analyzed the data. SH Lee designed the study and illustrated figures. HJ Shin designed the study, collected the data and wrote the manuscript. AG Lee revised the manuscript.

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Data availability Raw data table

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Code availability Not applicable

Ethics approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. The study protocol was approved by the Institutional Review Board (IRB) of Konkuk University Medical Center (registration number: KUH1100071).

Consent to participate Informed consent was obtained from all individual participants included in the study.

References

1. Metz HS, Cohen GH (1984) Quantitative forced duction measurement. Am Orthop J 34:28–38
2. Kushner BJ, Vrabec M (1987) Theoretical effects of surgery on length tension relationships in extraocular muscles. J Pediatr Ophthalmol Strabismus 24:126–131
3. Vishwanath M, Ansons A (2009) Midoperative forced duction test as a guide to titrate the surgery for consecutive exotropia. Strabismus. 17:41–44
4. Kim S, Yang HK, Hwang JM (2018) Surgical outcomes of unilateral recession and resection in intermittent exotropia according to forced duction test results. PLoS One 13:e0200741
5. Jung JH, Holmes JM (2015) Quantitative intraoperative torsional forced duction test. Ophthalmology. 122:1932–1938
6. Metz HS (1976) Forced duction, active force generation, and saccadic velocity tests. Int Ophthalmol Clin 16:47–73
7. Scott AB, Collins CC, O'Meara DM (1972) A forceps to measure strabismus forces. Arch Ophthalmol 88:330–333
8. Collins CC, O'Meara D, Scott AB (1975) Muscle tension during unrestrained human eye movements. J Physiol 245:351–369
9. Collins CC, Carlson MR, Scott AB, Jampolsky A (1981) Extraocular muscle forces in normal human subjects. Invest Ophthalmol Vis Sci 20:652–664
10. Lennerstrand G, Schiavi C, Tian S, Benassi M, Campos EC (2006) Isometric force measured in human horizontal eye muscles attached to or detached from the globe. Graefes Arch Clin Exp Ophthalmol 244:539–544
11. Rosenbaum AL, Myer JH (1980) New instrument for the quantitative determination of passive forced traction. Ophthalmology. 87: 158–163
12. Bekerman I, Gottlieb P, Vaiman M (2014) Variations in eyeball diameters of the healthy adults. J Ophthalmol 2014:503645
13. Lim HW, Lee DE, Lee JW, Kang MH, Seong M, Cho HY, Oh JE, Oh SY (2014) Clinical measurement of the angle of ocular movements in the nine cardinal positions of gaze. Ophthalmology. 121: 870–876
14. Stephens KF, Reinecke RD (1967) Quantitative forced duction. Trans Am Acad Ophthalmol Otolaryngo 71:324–329
15. Schillinger RJ (1966) The prevention of over-correction and under-correction in horizontal strabismus surgery. J Pediatr Ophthalmol Strabismus 3:38–41
16. Scott AB (1971) Extraocular muscle forces in strabismus. In: Bach-Y-Rita P, Collins CC, Hyde JE (eds) The Control of Eye Movements. Academic Press, New York, pp 327–342
17. Gao Z, Guo H, Chen W (2014) Initial tension of the human extraocular muscles in the primary eye position. J Theor Biol 353:78–83
18. Hertle RW, Granet DB, Zylan S (1993) The intraoperative oculocardiac reflex as a predictor of postoperative vaso-vagal responses during adjustable suture surgery. J Pediatr Ophthalmol Strabismus 30:306–311
19. Sainsbury P, Gibson JB (1954) Symptoms of anxiety and tension and the accompanying physiological changes in the muscle system. J Neurol Neurosurg Psychiatry 17:216–224
20. Rosenbaum AL, Santiago AP (1999) Clinical strabismus management: principles and surgical techniques. Saunders, Philadelphia, pp 39–41
21. Kallestad KM, Hebert SL, McDonald AA, Daniel ML, Cu SR, McLoon LK (2011) Sparing of extraocular muscle in aging and muscular dystrophies: a myogenic precursor cell hypothesis. Exp Cell Res 317:873–885